POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

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Chapter 2: Differential of a function

ANALYSIS 1

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Differential of Function

Definition

Let $X \subseteq \mathbb{R}, Y \subseteq \mathbb{R}$, a mapping

$$\begin{aligned} f : X &\to Y \\ x &\mapsto y = f(x) \end{aligned}$$

is called a *function*, the set X is *domain* of f. As usual, x is called *variable*, y is *function*. The value of $y = f(x)$ is a set of the form

$$\{f(x) : \ x \in X\}.$$

Even and odd functions

- Function $y = f(x)$ is *even* on the domain $D \Leftrightarrow \begin{cases} x \in D \Rightarrow -x \in D, \\ f(-x) = f(x) \quad \forall x \in D. \end{cases}$

- Function $y = f(x)$ is *odd* on the domain $D \Leftrightarrow \begin{cases} x \in D \Rightarrow -x \in D, \\ f(-x) = -f(x) \quad \forall x \in D \end{cases}$

Example as: $f_1(x) = |1 - x| + |1 + x|$ is even on $\mathbb{R}$ và $f_2(x) = x^4 \sin x$ is odd on $\mathbb{R}$.

Function $y = f(x)$ is called *periodic with period* T on the domain D, if there exists $T > 0$ such that

$$\begin{cases} x \in D \Rightarrow x + T \in D, \\ T \text{ is the smallest positive number such that } f(x+T) = f(x) \quad \forall x \in D. \end{cases}$$

Then, T is called *period* of f. Example as, the function $y = \sin(ax + b)$ where $a > 0$ is periodic with period $T = \frac{2\pi}{a}$ on $\mathbb{R}$.

Increasing and Decreasing Functions

+ Function $y = f(x)$ is called *increasing* on the domain D, if

$$\forall x_1, x_2 \in D, x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2).$$

+ Function $y = f(x)$ is called *strictly increasing* on D, if

$$\forall x_1, x_2 \in D, x_1 < x_2 \Rightarrow f(x_1) < f(x_2).$$

+ Function $y = f(x)$ is *decreasing* on D, if

$$\forall x_1, x_2 \in D, x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2).$$

+ Function $y = f(x)$ is *strictly decreasing* on D, if

$$\forall x_1, x_2 \in D, x_1 < x_2 \Rightarrow f(x_1) > f(x_2).$$

Bounded Functions

+ Function $y = f(x)$ is called *upper bounded* on D, if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \quad \forall x \in D.$$

+ Function $y = f(x)$ is called *lower bounded* on D, if exists $m \in \mathbb{R}$ such that

$$f(x) \geq m \quad \forall x \in D.$$

+ Function $y = f(x)$ is called *bounded* on D, if f is upper and power bounded on D.

Infimum and Supremum

+ Number m is called *infimum* of a function f on the domain D, denoted by $m = \inf_{x \in D} f(x)$, if and only if m is the biggest number such that

$$f(x) \geq m \quad \forall x \in D.$$

+ Number M is called *supremum* of f on D, denoted by

$$M = \sup_{x \in D} f(x),$$

if M is the smallest number such that

$$f(x) \leq M \quad \forall x \in D.$$

Infimum and Supremum (continuity)

Corollary

Number M is called the greatest lower bound of $f(x)$ trên D, if

$$\begin{cases} M \geq f(x) \quad \forall x \in D, \\ \forall \epsilon > 0 \Rightarrow \exists x_0 \in D \quad sao\ cho \quad M < f(x_0) + \epsilon. \end{cases}$$

Number m is the least upper bound of $f(x)$ on D, if

$$\begin{cases} m \leq f(x) \quad \forall x \in D, \\ \forall \epsilon > 0 \Rightarrow \exists x_0 \in D \quad sao\ cho \quad m > f(x_0) + \epsilon. \end{cases}$$

Let $X, Y \subseteq \mathbb{R}$ and an bijection $f : X \to Y$. Then, the inverse mapping $f^{-1} : Y \to X$ is also a function $y = f^{-1}(x)$ which is called *inverse function* of $y = f(x)$. Therefore,

$$y = f^{-1}(x) \Leftrightarrow x = f(y).$$

Example as, function $y = a^x$ where $a > 0, a \neq 1$ has an inverse function that is $y = log_a x$ where $x \in (0; +\infty)$. About their graph, the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$ is symmetric via the line $y = x$ on (Oxy).

a) *Function* $y = \arcsin x$

$$y = \arcsin x \Leftrightarrow \begin{cases} x = \sin y \\ y \in [-\frac{\pi}{2}; \frac{\pi}{2}]. \end{cases}$$

Thus, the domain of $y = \arcsin x$ là $D = [-1; 1]$ and the set of values is $D_f = [-\frac{\pi}{2}; \frac{\pi}{2}]$.

Example

Evaluate $\arcsin \frac{1}{2}$.

Some Important Inverse Functions (continuity)

b) *Function* $y = \arccos x$

$$y = \arccos x \Leftrightarrow \begin{cases} x = \cos y \\ y \in [0; \pi]. \end{cases}$$

The domain of $y = \arccos x$ is $D = [-1; 1]$ and the set of values is $D_f = [0; \pi]$.

Example

Show that

$$\arcsin x + \arccos x = \frac{\pi}{2} \quad \forall x \in [-1; 1].$$

c) *Function* $y = \arctan x$

$$y = \arctan x \Leftrightarrow \begin{cases} x = \tan y \\ y \in (-\frac{\pi}{2}; \frac{\pi}{2}). \end{cases}$$

The domain $y = \arctan x$ is $D = \mathbb{R}$ and the set of values $D_f = (-\frac{\pi}{2}; \frac{\pi}{2})$.

Example

Calculate $\arctan \sqrt{3}$.

d) *Function $y = arccotx$*

$$y = \text{arccot}x \Leftrightarrow \begin{cases} x = \cot y \\ y \in (0; \pi). \end{cases}$$

The domain $y = \text{arccot}x$: $D = \mathbb{R}$ and the set of values: $D_f = (0; \pi)$.

Example

Prove
$$\arctan x + \text{arccot } x = \frac{\pi}{2}.$$

e) *Hypebolic functions*
Sinhypebolic: $y = \sinh(x)$, it is defined by

$$\sinh(x) = \frac{1}{2}(e^x - e^{-x}).$$

Cosinhypebolic: $y = \cosh(x)$, it is given in the form:

$$\cosh(x) = \frac{1}{2}(e^x + e^{-x}).$$

Example

Let $x, y \in \mathbb{R}$. Prove that:
$\sinh(x+y) = \sinh(x).\cosh(y) + \cosh(x).\sinh(y)$,
$\sinh(x-y) = \sinh(x).\cosh(y) - \cosh(x).\sinh(y)$,
$\cosh(x+y) = \cosh(x).\cosh(y) + \sinh(x).\sinh(y)$,
$\cosh(x-y) = \cosh(x).\cosh(y) - \sinh(x).\sinh(y)$,
$\cosh^2(x) - \sinh^2(x) = 1$,
$\cosh^2(x) + \sinh^2(x) = \cosh(2x)$.

Definition

Let a function $y = f(x)$ and $x_0 \in \mathbb{R}$. The function $f(x)$ has the *limit* A (definite) as x approaches x_0, denoted by $\lim\limits_{x \to x_0} f(x) = A$, if

$$\forall \epsilon > 0,\ \exists \delta > 0 \ \text{ such that } \ 0 < |x - x_0| < \delta \ \ \forall x \in (a, b) \Rightarrow |f(x) - A| < \epsilon.$$

Then, $f(x)$ has the *left limit* $A \in \mathbb{R}$ as x approaches x_0, denoted by $\lim\limits_{x \to x_0^-} f(x) = A$, if

$$\forall \epsilon > 0,\ \exists \delta > 0 \ \text{ such that } 0 > x - x_0 > -\delta \Rightarrow |f(x) - A| < \epsilon.$$

Function $f(x)$ has the *right limit* $A \in \mathbb{R}$ as x approaches x_0, denoted by $\lim\limits_{x \to x_0^+} f(x) = A$, if

$$\forall \epsilon > 0,\ \exists \delta > 0 \ \text{ such that } \ 0 < x - x_0 < \delta \ \ \forall x \in (a, b) \Rightarrow |f(x) - A| < \epsilon.$$

Definition (continuity)

Remark

$$\lim_{x\to x_0} f(x) = A \Leftrightarrow \begin{cases} \lim\limits_{x\to x_0^-} f(x) = A \\ \lim\limits_{x\to x_0^+} f(x) = A. \end{cases}$$

- The limit $\lim\limits_{x\to +\infty} f(x) = A$, if

$$\forall \epsilon > 0,\ \exists x_0 \ \text{ sao cho } \forall x > x_0 \Rightarrow |f(x) - A| < \epsilon.$$

- The limit $\lim\limits_{x\to -\infty} f(x) = A$, if

$$\forall \epsilon > 0,\ \exists x_0 \ \text{ sao cho } \forall x < x_0 \Rightarrow |f(x) - A| < \epsilon.$$

Example

Using the definition to prove that

$$\lim_{x\to 1} \frac{x+1}{2x-1} = 2.$$

Properties of the limit

- *Boundedness*

Theorem

If $\lim\limits_{x\to x_0} f(x) = A \in \mathbb{R}$, *then* $f(x)$ *is bonded on a neighbourhood of* x_0.

- *Order*

Theorem

i) Suppose that there exist definite the limits $\lim\limits_{x\to x_0} f(x)$, $\lim\limits_{x\to x_0} g(x)$. *If* $f(x) \le g(x) \quad \forall x \in$
$(x_0 - \delta, x_0 + \delta)$, x *can be not equal to* x_0 *then* $\lim\limits_{x\to x_0} f(x) \le \lim\limits_{x\to x_0} g(x)$.
ii) If $\lim\limits_{x\to x_0} f(x) > m$, *then there exists* $\delta > 0$ *such that*
$f(x) > m \quad \forall x_0 \ne x \in (x_0 - \delta, x_0 + \delta)$.
iii) If $f(x) \le g(x) \quad \forall x \in (x_0 - \delta, x_0 + \delta)$ *và* $\lim\limits_{x\to x_0} f(x) = +\infty$, *then* $\lim\limits_{x\to x_0} g(x) = +\infty$.

Number $e \approx 2,71718...$

Definition

(The first form)

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Example

Using the definition to calculate

$$I = \lim_{x\to\infty}\left(\frac{x^2+x+1}{x^2-3x-2}\right)^{\frac{x^3-3x}{1-x^2}}$$

Formula

(The second form)

$$\lim_{y\to 0}\left(1+y\right)^{\frac{1}{y}} = e.$$

Example

Let $a \neq 0, b \neq 0$. Using this formula to calculate

$$J = \lim_{x\to 0}(\cos ax)^{\frac{1}{x\sin bx}}.$$

Formula

(The third form)

$$\lim_{y \to 0} \frac{e^y - 1}{y} = 1.$$

Example

Evaluate

$$J = \lim_{x \to 0} \frac{2^{x^2} - \cos 2x}{1 - \sqrt{1 + 2x^2}}.$$

Formula

(The fourth form)

$$\lim_{y\to 0} \frac{\ln(1+y)}{y} = 1.$$

Example

Evaluate

$$K = \lim_{x\to 0} \log_{1-x\sin 3x}(\cos 4x + x^2).$$

Properties

Let definite limits $\lim\limits_{x \to x_0} f(x) = A$ and $\lim\limits_{x \to x_0} g(x) = B$ $(x_0 \in \bar{\mathbb{R}} = [-\infty, +\infty])$. Then, we have follows

a) $\lim\limits_{x \to x_0} [f(x) + g(x)] = \lim\limits_{x \to x_0} f(x) + \lim\limits_{x \to x_0} g(x)$.

b) $\lim\limits_{x \to x_0} [f(x) - g(x)] = \lim\limits_{x \to x_0} f(x) - \lim\limits_{x \to x_0} g(x)$.

c) $\lim\limits_{x \to x_0} [k.f(x)] = k \lim\limits_{x \to x_0} f(x)$, for all constant $k \in \mathbb{R}$.

d) $\lim\limits_{x \to x_0} [f(x).g(x)] = \lim\limits_{x \to x_0} f(x). \lim\limits_{x \to x_0} g(x)$.

e) $\lim\limits_{x \to x_0} \dfrac{f(x)}{g(x)} = \dfrac{\lim\limits_{x \to x_0} f(x)}{\lim\limits_{x \to x_0} g(x)}$, where $B \neq 0$.

Definition

Let $y = f(x)$ have the domain D and $x_0 \in D$. The function $f(x)$ is called *continuous* at x_0, if $\lim\limits_{x \to x_0} f(x) = f(x_0)$. It means that

$$\forall \epsilon > 0,\ \exists \delta > 0, \forall x \in D : 0 < |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \epsilon.$$

The function $f(x)$ is called *left continuous* at $x_0 \in D$, if $\lim\limits_{x \to x_0^-} f(x) = f(x_0)$, i.e.,

$$\forall \epsilon > 0,\ \exists \delta > 0, \forall x \in D : 0 > x - x_0 > -\delta \Rightarrow |f(x) - f(x_0)| < \epsilon.$$

The function $f(x)$ is called *right continuous* at $x_0 \in D$, if $\lim\limits_{x \to x_0^+} f(x) = f(x_0)$.

Remark

A function $f(x)$ is continuous at x_0 if and only if $f(x)$ is left and right continuous at x_0.

Discontinuous point

A function $f(x)$ is called *discontinuous* at x_0, if $f(x)$ is not continuous x_0. Then, the point x_0 is called *discontinuous point* of $f(x)$. If x_0 is a discontinuous point of $f(x)$ and there exists the definite limits $\lim\limits_{x \to x_0^+} f(x) = f(x_0^+)$, $\lim\limits_{x \to x_0^-} f(x) = f(x_0^-)$, then $h = |f(x_0^+) - f(x_0^-)|$ is called *step-length* of the function $f(x)$ at x_0.

A function $f(x)$ is continuous on an interval if it is continuous at every number in the interval. (If $f(x)$ is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

Example

Find a, b such that f is continuous on $\mathbb{R}$

$$f(x) = \begin{cases} ax^2 - bx + 1 & \text{if } x < 1, \\ 2a - b & \text{if } 1 \leq x \leq 2, \\ bx + a - 1 & \text{if } x > 2. \end{cases}$$

Properties

Let $f(x), g(x)$ be continuous on D. Then, the functions $f(x)+g(x), f(x).g(x), kg(x)$ (where k : constant), $|f(x)|$, $\frac{f(x)}{g(x)}$ (where $g(x) \neq 0$) are continuous on D.

Corollary

+ Polynomial function $y = P_n(x)$ is continuous on $\mathbb{R}$.
+ Rational function $y = \frac{P_n(x)}{Q_m(x)}$ is continuous for all $x_0 \in \mathbb{R}$ such that $Q_m(x_0) = 0$.
+ Trigonometric function, exponential function, root functions, logarithmic function, hypebolic function or inverse function are continuous on their domain.

Example

Let $f(x)$ and $g(x)$ be continuous on D. Prove that $\min\{f(x), g(x)\}$ and $\max\{f(x), g(x)\}$ are also continuous on D.

Properties (continuity)

Theorem

(The Intermediate Value Theorem) Suppose that $f(x)$ is continuous on the closed interval $[a, b]$ and let α be any number between $f(a)$ and $f(b)$. Then, there exists a number $x_0 \in [a, b]$ such that $f(x_0) = \alpha$.

Example

Prove that the equation has any real solution

$$4x^3 - 6x^2 + 3x - 2 = 0.$$

Definition

Let a function $f(x)$ define on interval (a, b) and $x_0 \in (a, b)$.

+ The derivative of $f(x)$ at x_0, denoted by $f'(x_0)$, is

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta_x \to 0} \frac{f(x_0 + \Delta_x) - f(x_0)}{\Delta_x}.$$

+ The right derivative of $f(x)$ at x_0, denoted by $f'(x_0^+)$, is

$$f'(x_0^+) = \lim_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta_x \to 0^+} \frac{f(x_0 + \Delta_x) - f(x_0)}{\Delta_x}.$$

+ The left derivative of $f(x)$ at x_0, denoted by $f'(x_0^-)$, is

$$f'(x_0^-) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{\Delta_x \to 0^-} \frac{f(x_0 + \Delta_x) - f(x_0)}{\Delta_x}.$$

+ The function $f(x)$ is differentiable at x_0 if $f'(x_0)$ exists. It is differentiable on an open interval $(a, b), (a, +\infty), (-\infty, b), (-\infty, +\infty)$, if it is differentiable at every number in the interval.

Relations

A function $f(x)$ is differentiable at x_0 if and only if $f(x)$ has the right derivative $f_+(x_0)$ and the left derivative $f'_+(x_0)$ such that $f'_-(x_0) = f'_+(x_0)$.

Example

Prove that the function $f(x) = x|x-1|$ exist $f'_-(1)$ and $f'_+(1)$, but it does not exists $f'(1)$.

Formula of derivatives

Let $f(x)$ and $g(x)$ be differentiable on (a, b). Then

+ $\Big(f(x) + g(x)\Big)' = f'(x) + g'(x)$.

+ $\Big(f(x) - g(x)\Big)' = f'(x) - g'(x)$.

+ $\Big(f(x).g(x)\Big)' = f'(x).g(x) + f(x).g'(x)$.

+ $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x).g(x) - f(x).g'(x)}{g^2(x)}$.

+ $\Big(kf(x)\Big)' = kf'(x)$.

+ The composite function: $y = f(u), u = u(x)$. Then, $f'_x = f'_u.u'_x$.

+ The inverse function: $y'_x = \frac{1}{x'_y}$.

Derivative of popular functions

$y = C \ (constant) \Rightarrow y' = 0 \ \ \forall x \in \mathbb{R}.$
$y = x^\alpha (\alpha \in \mathbb{R}) \Rightarrow y' = \alpha x^{\alpha - 1} \ \ \forall x \in \mathbb{R}.$
$y = \sin x \Rightarrow y' = \cos x \ \ \forall x \in \mathbb{R}.$
$y = \cos x \Rightarrow y' = -\sin x \ \ \forall x \in \mathbb{R}.$
$y = \tan x \Rightarrow y' = \frac{1}{\cos^2 x} \ \ \forall x \in \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : \ k \in \mathbb{Z}\}.$
$y = \cot x \Rightarrow y' = -\frac{1}{\sin^2 x} \ \ \forall x \in \mathbb{R} \setminus \{k\pi : \ k \in \mathbb{Z}\}.$
$y = e^x \Rightarrow y' = e^x \ \ \forall x \in \mathbb{R}.$
$y = a^x \ (\text{where } 1 \neq a > 0) \Rightarrow y' = a^x \ln a \ \ \forall x \in \mathbb{R}.$
$y = \log_a x \ (\text{where } 1 \neq a > 0) \Rightarrow y' = \frac{1}{x \ln a} \ \ \forall x \in \mathbb{R}^* = \{x \in \mathbb{R} : \ x > 0\}.$
$y = sinh \ x \Rightarrow y' = coshx \ \ \forall x \in \mathbb{R}.$
$y = cosh \ x \Rightarrow y' = sinhx \ \ \forall x \in \mathbb{R}.$
$y = tanh \ x \Rightarrow y' = \frac{1}{cosh^2 x} \ \ \forall x \in \mathbb{R}.$
$y = coth \ x \Rightarrow y' = \frac{-1}{sinh^2 x} \ \ \forall x \in \mathbb{R} \setminus \{0\}.$

Some inverse functions

$+\ y = \arcsin x \Rightarrow y' = \frac{1}{\sqrt{1-x^2}} \quad \forall x \in (-1, 1).$
$+\ y = \arccos x \Rightarrow y' = \frac{-1}{\sqrt{1-x^2}} \quad \forall x \in (-1, 1).$
$+\ y = \arctan x \Rightarrow y' = \frac{1}{1+x^2} \quad \forall x \in \mathbb{R}.$
$+\ y = \text{arccot } x \Rightarrow y' = \frac{-1}{1+x^2} \quad \forall x \in \mathbb{R}.$

Differential

Let $f(x)$ be differentiable at x, be the definition

$$\lim_{\Delta_x \to 0} \frac{f(x+\Delta_x) - f(x)}{\Delta_x} = f'(x).$$

It means that

$$f(x+\Delta_x) - f(x) = f'(x)\Delta_x + o(\Delta_x).$$

Product $f'(x)\Delta_x$ is called *differential* of $f(x)$ at x, denoted by $df(x)$. Then, we get

$$df(x) = f'(x)dx.$$

$+ \, d\Big(f(x) + g(x)\Big) = d\Big(f(x)\Big) + d\Big(g(x)\Big).$

$+ \, d\Big(f(x) - g(x)\Big) = d\Big(f(x)\Big) - d\Big(g(x)\Big).$

$+ \, d\Big(f(x).g(x)\Big) = g(x)d\Big(f(x)\Big) + f(x)d\Big(g(x)\Big).$

$+ \, d\Big(\frac{f(x)}{g(x)}\Big) = \frac{g(x)d\Big(f(x)\Big) - f(x)d\Big(g(x)\Big)}{g^2(x)}.$

Derivative with order n

+

If $f(x)$ is differentiable at x, then $f'(x)$ is called *the first derivative* of $f(x)$.
+ If $f'(x)$ is differentiable at x, then $f"(x)$ is called *the second derivative* of $f(x)$.
+ If $f"(x)$ is differentiable at x, then $f'''(x)$ is called *the third derivative* of $f(x)$.
+ By mathematical induction, if $f^{(n-1)}(x)$ is differentiable at x, then $f^{(n)}(x)$ is called *the n^{th} derivative* of $f(x)$. Then, we also have *the n^{th} differential* of $f(x)$ as follows

$$d^n f(x) = d(d^{n-1} f(x)) = f^{(n)}(x) dx^n.$$

Some formula of the n^{th} derivative

$+ \Big(f(x)+g(x)\Big)^{(n)} = \Big(f(x)\Big)^{(n)} + \Big(g(x)\Big)^{(n)}.$

$+ \Big(f(x)-g(x)\Big)^{(n)} = \Big(f(x)\Big)^{(n)} - \Big(g(x)\Big)^{(n)}.$

$+ \Big(kf(x)\Big)^{(n)} = k\Big(f(x)\Big)^{(n)}$ với $k = const.$

$+ (uv)^{(n)} = \sum_{k=0}^{n} C_n^k u^{(k)} v^{(n-k)}$ where

$u = u(x), v = v(x), u^{(0)} = u, v^{(0)} = v.$

Popular forms

Form 1:

Evaluate the n^{th} derivative

$$f(x) = \frac{1}{ax+b} \quad \text{where} \quad a, b : const.$$

Example

By using Form 1 to calculate the n^{th} derivative

$$f(x) = \frac{x^2+1}{(x-1)^3(x+3)}.$$

Form 2:

Evaluate the n^{th} derivative

$$f(x) = \sin(ax+b) \ \text{ với } a, b : const.$$

Example

By using Form 1 to calculate the n^{th} derivative

$$f(x) = \sin^3 x \cos^2 2x.$$

Example

By using Form 1 to calculate the n^{th} derivative

$$f(x) = (x^2+3x-1)\sin 3x.$$

Differential of Function

Maximum and Minimum Values

Definition

A function $f(x)$ has an *absolute maximum* (or *global maximum*) at $x_0 \in D$ if $f(x_0) \geq f(x)$ for all $x \in D$, where D is the domain of $f(x)$. The number $f(x_0)$ is called the *maximum* value of $f(x)$ on D. Similarly, $f(x)$ has an *absolute minimum* at $x_1 \in D$ if $f(x_1) \leq f(x)$ for all x in D and the number $f(x_1)$ is called the *minimum value* of $f(x)$ on D. The maximum and minimum values of are called the *extreme* values of $f(x)$.

Definition

A function $f(x)$ has an *local maximum* at $x_0 \in D$ if $f(x_0) \geq f(x)$ for all $x \in D$ and x is near x_0, where D is the domain of $f(x)$. Similarly, $f(x)$ has an *local minimum* at $x_1 \in D$ if $f(x_1) \leq f(x)$ for all x in D and x is near x_1.

Fermat's theorem (Pierre de Fermat: 17.8.1601-12.1.1665)

Theorem

If $f(x)$ *is differentiable at* x_0 *and obtains the extreme at* x_0*, then* $f'(x_0) = 0$.

Rolle's theorem (Michel Rolle: 21.4.1652-8.11.1719)

Theorem

Let a function $f(x)$ be continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$. Then, there exists $c \in (a, b)$ such that $f'(c) = 0$.

Example

Let a function $f(x)$ be continuous on $[a, b]$, differentiable on (a, b) and

$$\begin{cases} f(a) = f(b) = 0 \\ f'(a^+) > 0 \\ f'(b^-) < 0. \end{cases}$$

Prove that $\exists\ c_1, c_2, c_3 \in (a, b)$ such that

$$c_1 < c_2 < c_3, f(c_2) = 0, f'(c_1) = f'(c_3) = 0.$$

Fermat's theorem (Pierre de Fermat: 17.8.1601-12.1.1665)

Theorem

If $f(x)$ is differentiable at x_0 and obtains the extreme at x_0, then $f'(x_0) = 0$.

Lagrange's theorem (Joseph Louis Lagrange: 25.1.1736-10.4.1813)

Theorem

Let a function $f(x)$ be continuous on $[a, b]$, differentiable on (a, b). Then, there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Lagrange's theorem (Joseph Louis Lagrange: 25.1.1736-10.4.1813)

Example

Let a function $f(x)$ be continuous on $[a, b]$, differentiable on (a, b) excepting n points. Prove that there exist $n+1$ positive numbers $\alpha_1, \alpha_2, ..., \alpha_{n+1}$ and $n+1$ numbers c_i $(i = 1, 2, ..., n+1)$ such that

$$\begin{cases} \sum_{i=1}^{n+1} \alpha_i = 1 \\ a < c_1 < c_2 < ... < c_{n+1} < b \\ \frac{f(b)-f(a)}{b-a} = \sum_{i=1}^{n+1} \alpha_i f'(c_i). \end{cases}$$

Cauchy's theorem

Theorem

Let two functions $f(x), g(x)$ be continuous on $[a, b]$, differentiable on (a, b) such that $g'(x) \neq 0$ for all $x \in (a, b)$. Then, there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Definition

Let a function $f(x)$ be differentiable with $(n+1)^{th}-$degree in a neighbourhood of point x_0. Then, an $n^{th}-$degree polynomial $P_n(x)$ is called $n^{th}-$*degree Taylor polynomial* of $f(x)$ centered at x_0 if and only if

$$\begin{cases} P_n(x_0) = f(x_0) \\ P_n'(x_0) = f'(x_0) \\ ... \\ P_n^{(n)}(x_0) = f^{(n)}(x_0). \end{cases}$$

In the special case, if $x_0 = 0$ then $P_n(x)$ is called $n^{th}-$*degree Maclaurin polynomial* of $f(x)$.

Taylor Polynomial (continuity)

Let a function $f(x)$ be differentiable with $(n+1)^{th}-$degree in a neighbourhood of point x_0.

Theorem

If $P_n(x)$ is an $n^{th}-$degree Taylor polynomial of $f(x)$ centered at x_0, then

$$P_n(x) = f(x_0)+\frac{1}{1!}f'(x_0)(x-x_0)^1+\frac{1}{2!}f''(x_0)(x-x_0)^2+...+\frac{1}{n!}f^{(n)}(x_0)(x-x_0)$$

The $n^{th}-$degree Taylor redundance

Let $P_n(x)$ is an $n^{th}-$degree Taylor polynomial of $f(x)$ centered at x_0. Then, *the $n^{th}-$degree Taylor redundance* of $f(x)$, denoted by $R_n(x)$, is defined in the form

$$R_n(x) = f(x) - P_n(x).$$

Theorem

If $f(x)$ is differentiable with $(n+1)^{th}-$degree in a neighbourhood $(x_0 - \delta, x_0 + \delta)$ and $x \in (x_0 - \delta, x_0 + \delta)$, then there exists c belonging to between x and x_0, i.e., $c = x_0 + \lambda(x - x_0)$ where $\lambda \in (0, 1)$ such that

$$R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}.$$

The $n^{th}-$degree Taylor formula

Let $f(x)$ be differentiable with $(n+1)^{th}-$degree in a neighbourhood $(x_0-\delta, x_0+\delta)$ and $x \in (x_0-\delta, x_0+\delta)$, then *the $n^{th}-$degree Taylor formula* is formulated by

$$f(x) = f(x_0)+\frac{f'(x_0)}{1!}(x-x_0)^1+\frac{f''(x_0)}{2!}(x-x_0)^2+...+\frac{f^{(n)}(x_0)}{n!}(x-x_0)^n+R_n($$

where
$\lambda \in (0,1), c = x_0+\lambda(x-x_0), R_n(x_0) = \frac{1}{(n+1)!}f^{(n+1)}(c)(x-x_0)^{n+1}.$
In the special case $x_0 = 0$, the above formula is called *the $n^{th}-$degree Maclaurin formula.*

Example

Present the $n^{th}-$degree Taylor formula centered at x_0 of the following functions:

$$(a) f(x) = e^x \text{ at } x_0 = 0.$$
$$(b) f(x) = \sin x \text{ at } x_0 = 0.$$
$$(c) f(x) = \cos x \text{ at } x_0 = 0.$$
$$(d) f(x) = \ln(1+x) \text{ at } x_0 = 0.$$
$$(e) f(x) = \ln(2x - x^2 + 3) \text{ at } x_0 = 2.$$

L'Hospital Rule

Theorem

Let two functions $f(x)$ and $g(x)$ define in a neighbourhood $(x_0 - \delta, x_0 + \delta)$ of point x_0 (except x_0). Suppose that

i) $\lim\limits_{x \to x_0} f(x) = \lim\limits_{x \to x_0} g(x) = 0$

ii) $g'(x) \neq 0$ *for all* $x \in (x_0 - \delta, x_0 + \delta)$ *(except x_0).*

iii) There exists $\lim\limits_{x \to x_0} \frac{f'(x)}{g'(x)} = A$.

Then, there exists the limit $\lim\limits_{x \to x_0} \frac{f(x)}{g(x)}$ *(form: $\frac{0}{0}, \frac{\infty}{\infty}$) and*

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)}.$$

Using L'Hospital to evaluate the limits:

$$I_1 = \lim_{x\to 1} \frac{x^x - 1}{\ln x + x - 1}.$$
$$I_2 = \lim_{x\to 0} \frac{1}{x}\left(\frac{1}{\tan x} - \frac{1}{\tan x}\right).$$
$$I_3 = \lim_{x\to +\infty} \left(x + \sqrt{x^2+1}\right)^{\frac{1}{\ln x}}.$$